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Research Note

Benefits from optimal route planning based on *B-patterns*

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B-patterns are algorithmically-computed sequences of field-work tracks for agricultural vehicles completely covering an area that do not follow any pre-determined standard motif, as in the conventional field-work patterns, but are a result of an optimisation process under one or more selected criteria. Here, an initial approach for the assessment of the benefits derived from the implementation of *B-patterns* is presented. This approach is based on the comparison of *B-patterns* against conventional field work patters, by simulating the operational performance of the different patterns on template fields for different combinations of field and machinery characteristics. The assessment of the benefits of *B-patterns* showed reductions in non-working distance up to 58.65% and increases in the area capacity, i.e. the processed area per time unit, were up to 19.23% when compared with different types of conventional field-work patterns.

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1. Introduction

Conventionally, in order to cover a field area, the routes followed by agricultural vehicles are formed as repetitions of standard motifs. The factor determining this practice is the human operator's lack of ability to precisely distinguish the succeeding field-work track to be followed from the current one. However, implementation of sensor systems as auxiliary systems for agricultural vehicles navigation (e.g., geographical positioning systems, and machine vision systems) has relaxed this constraining factor. This has provided the possibility of generating optimised field area coverage practises (Jin and Tang, 2010, 2011). Following this principle, a new type of optimal field-work patterns, *B-patterns*, have been recently introduced (Bochtis, 2008). *B-patterns* are defined as: *algorithmically-computed sequences of field-work tracks completely covering an area and that do not follow any pre-determined standard motif, but in contrast, are a result of an optimisation process under one or more selected criteria*. It should be noted that *B-patterns* do not generate any sub-field areas division but, in contrast, the generation of the sub-fields (when it is needed due to e.g. physical obstacles or complex field shapes) is a prerequisite for applying *B-patterns* methodology (de Bruin, Lerink, Klompe, van derWal & Heijting, 2009; Oksanen & Visala, 2009). The optimal sequences are the unique result of the optimisation approach on the specific combination of the mobile unit kinematics and dimensions, the operating width, the field shape, and the optimisation criterion-criteria. However, they can provide the optimal sequence of covering the sub-fields since this is an inherent part of the track sequence optimisation solution. Figure 1 demonstrates the optimal sequence of tracks in the case of two selected experimental fields. In

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principle, the optimisation process involves the expression of the field are coverage as the traversal of a weighted graph, and the problem of finding optimal traversal sequence or the field-work tracks is equivalent to finding the shortest tour in the weighted graph. The weight of the graph arcs could be based on one or more optimisation criteria, such as, total or non-working travelling distance, (e.g. Bochtis, Vougioukas & Griepentrog, 2009), total operational time (e.g. Spekken and de Bruin, 2012), a soil compaction measure (e.g. Bochtis, Sørensen & Green, 2012), etc. The underlying optimisation problem in a mathematical form is given in Appendix I.

B-patterns can be implemented in a range of field operations, for single or multiple-machinery systems, and for deterministic, stochastic, or dynamic operational characteristics, as it has been outlined in Bochtis and Sørensen (2009) were the different instances of the well-known combinatorial optimisation problem, the vehicle routing problem (VRP) have been employed for the generation of *B-patterns* for different types of field operations.

B-patterns have been implemented and tested in an autonomous agricultural vehicle (Bochtis et al. 2009) showing that, under the criterion of the minimized non-working distance this distance can be reduced significantly reaching up to 50% for the specific vehicle in a range of different operations. Here, an initial approach for the assessment of the benefits derived from the implementation of *B-patterns* is presented. This approach is based on the comparison of *B-patterns* against conventional field work patters consisting of repetitions of standard motifs, by simulating the operational performance of the different patterns on template fields for different combinations of field and machinery characteristics.

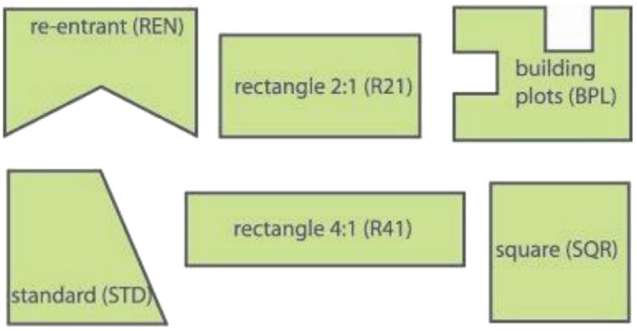


Fig. 2 – Template field shapes representing typical fields (adapted from Witney, 1995).

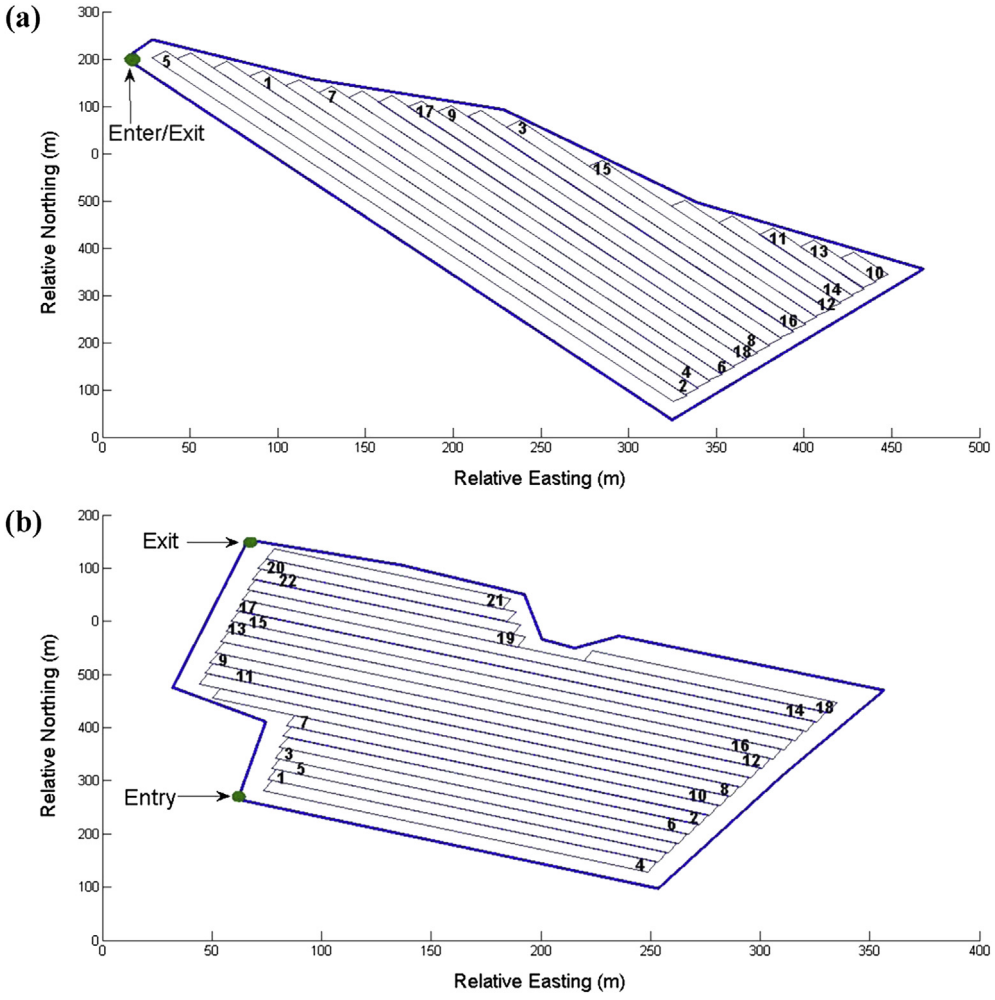


Fig. 1 – Optimal sequences under the criterion of minimising operational time for two selected experimental fields a) [56°29′35″N, 9°35′22″E], and b) [56°29′42″N, 9°35′37″E] (operating width: 10 m, minimum turning radius: 7.5 m).

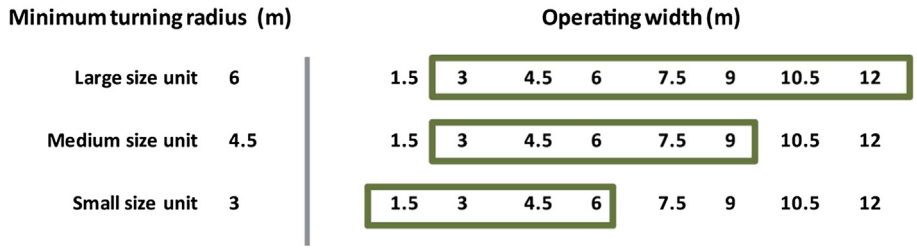


Fig. 3 – The combinations of minimum turning radius of the mobile unit and operating width of the implement employed in the assessment.

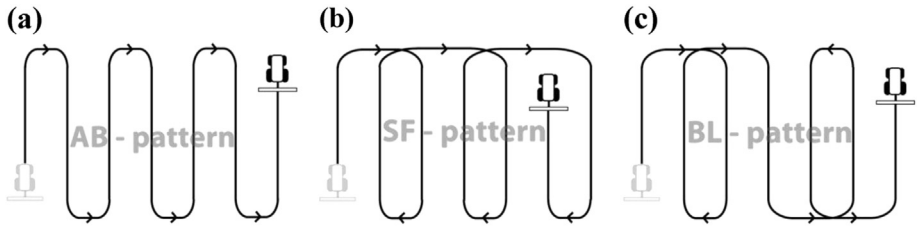


Fig. 4 – The three types of field-work patterns employed in the comparison (the depicted path-lines presents the sequence of the tracks and not the actual path which depends on the turning type).

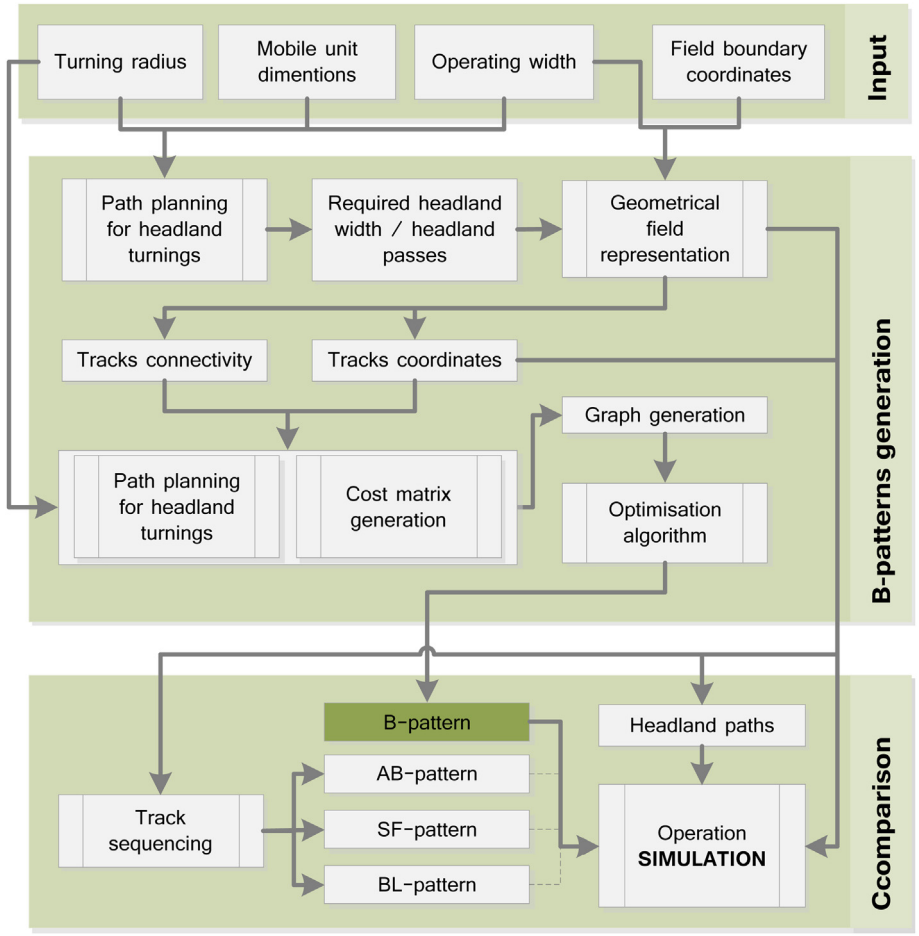


Fig. 5 – The architecture of the *B-patterns* generation and the subsequent comparison with the conventional patterns for each selected scenario.

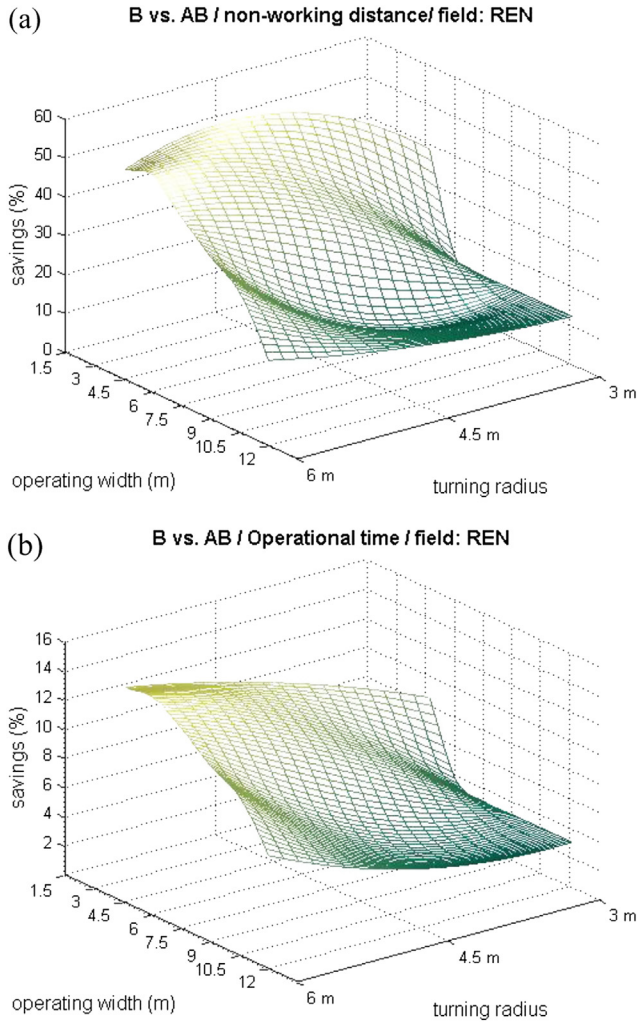


Fig. 6 – Savings in non-working distance (a) and operational time (b) from the implementation of *B-patterns* instead of AB-pattern in the case of field REN.

2. Assessment methodology

The assessment regards the savings in non-working travelled distance and the increases in area capacity (e.g. ha h^{-1}) as resulting from the implementation of the optimal field-work patterns. The assessment presented here does not regard coupled machines operations, where a primary unit has to be supported in-field by a service unit (e.g. the grain harvesting case). The assessment is based on 288 combinations derived from the consideration of six field shapes, three conventional fieldwork patterns, and sixteen different combinations of mobile unit kinematics and implement operating widths.

2.1. *Inputs*

2.1.1. *The test fields*

In order to investigate the influence of the field shape on the savings, a set of template fields of the same area (10 ha) and different shapes were selected as test fields (Fig. 2). The particular template field shapes are typical abstractions of real field shapes which have been used in a number of studies related to machinery management (Gunnarsson, Vagstrom & Hansson, 2008; Landers, 2000, p. 152; Sturrock, Cathie & Payne, 1977; Witney, 1995).

2.1.2. *The machinery*

In order to include different sizes of machinery three cases in terms of sizing and consequently in manoeuvrability (minimum turning radius) were considered. Specifically, a large sized unit with minimum turning radius of 6 m, a medium sized unit with minimum turning radius of 4.5 m, and a small sized unit with minimum turning radius of 3 m, were selected. Eight operating widths were considered ranging from 1.5 m to 12 m with an incremental step of 1.5 m. The examined combinations of turning radius and operating width are presented in Fig. 3. Only symmetric combinations were considered. The excluded combinations regard a) the case of large unit matched with very small operating width b) the case of small units matched with large operating widths, since a small sized unit is not able to provide the power for a large operating width, and c) the case where the operating width exceeds twice the minimum turning radius, since optimal solutions in that cases coincides with the ones where the operating width equally twice the minimum turning radius and consequently the savings are the same as in the later case.

2.1.3. *The conventional field-work patterns*

Three different field-work patterns were selected for the assessment. The specific patterns represent currently implemented functions in state-of-the-art auto-steering systems and include:

a. AB-pattern (Fig. 4a)
b. SF-pattern (Fig. 4b). "Skip and fill" patterns according to the terminology used in iTEC PRO (2007)
c. BL-patterns (Fig. 4c). "First turn skip" patterns according to the terminology used in iTEC PRO (2007).

The first example (AB-pattern), is the most common field-work pattern used in field operations (Fig. 4a). The other two have been selected in order to represent the cases where the consecutive tracks covered by the unit are non-adjacent. The mathematical descriptions of the above mentioned patterns are provided in Appendix II.

2.2. *Architecture*

The generic architecture of the assessment tool is presented in Fig. 5. The assessment tool combines the generation of the optimal field-work patterns, the simulation of the operation following the generated optimal pattern and following the three conventional patterns, and the subsequent comparison.

For the estimation of the length and shape of each individual headland turning a path planning module for non-holonomic systems in obstacle-free planes following the Ackerman steering kinematics for agricultural vehicles (Bochtis, 2008), was invoked. Using this module, in a first stage based on the implement's operating width, the unit's dimensions and turning radius, the required minimum

Table 1 – Savings statistics for combinations of field shapes and conventional pattern types.

		Non-working distance savings (%)				Area capacity savings (%)			
		Average	Max	Min	Field average	Average	Max	Min	Field average
SQR	AB	31.18	57.74	3.73	34.90	7.69	12.57	0.50	6.24
	SF	38.90	52.63	23.24		5.81	10.61	2.03	
	BL	34.62	53.68	16.85		5.21	11.41	1.38	
R21	AB	41.31	57.41	3.59	39.29	5.49	8.81	1.25	4.44
	SF	40.17	54.32	19.14		4.10	7.51	1.19	
	BL	36.38	51.88	14.92		3.74	8.02	0.85	
R41	AB	38.14	58.65	4.03	36.30	3.54	5.71	0.96	2.86
	SF	37.09	55.42	14.60		2.63	4.87	1.06	
	BL	33.67	56.45	10.03		2.41	5.21	0.68	
STD	AB	45.42	58.36	11.25	43.22	9.75	14.99	2.71	7.98
	SF	43.79	56.53	28.56		7.19	13.37	2.22	
	BL	40.45	56.06	20.48		6.98	13.87	1.51	
REN	AB	44.31	58.51	13.46	42.14	12.11	18.46	3.80	9.83
	SF	43.10	54.80	28.12		9.00	16.72	3.18	
	BL	39.01	55.35	20.67		8.37	17.17	2.22	
BPL	AB	44.46	58.44	14.75	42.37	12.35	19.23	4.01	10.05
	SF	43.50	54.53	29.13		9.34	15.96	4.25	
	BL	39.14	57.96	21.20		8.47	17.17	3.20	

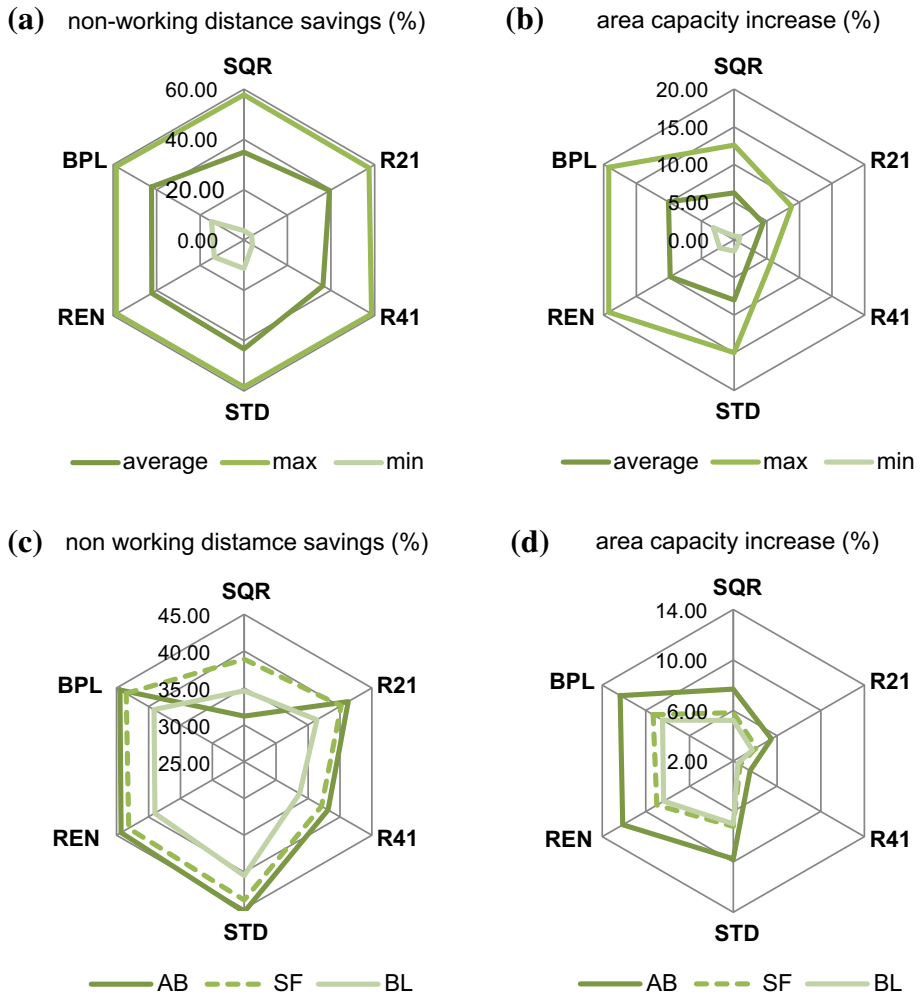


Fig. 7 – Overview of the relative importance of the relative parameters – (a) and (b) regard the selective case of comparing with *AB-patterns*, (c) and (d) regard the case of the average values in non-working distance and increases in area capacity, respectively.

headland width was estimated and the corresponding number of headland passes. Based on the number of headland passes and the coordinates of the field boundary the geometrical representation of the field was generated providing the field-work track coordinates and the tracks connectivity in the generated sub-fields area (only valid for the BPL field). In a second stage, the same path planning module was implemented in order to estimate the paths that connect each possible pair of tracks generating the cost matrix of the optimisation problem. The optimisation problem was then solved using an implementation of the Clarke–Wright savings algorithm (Clarke and Wright, 1964) and the optimal track sequence (*B-pattern*) was provided. Based on the geometrical representation of the field and the mathematical description of the conventional field-work patterns, the track sequences for the three cases of the conventional coverage were also generated. Then the traversal of all of the four field-work patterns was simulated. The traversal of the headland passes was common for all of the four cases.

For the estimation of the task times an average operating speed of 2 m $s^{-1}$was assumed in order to cover the majority of the field operations. A reduction factor of the operating speed of 90% and of 95% was assumed considered for the cases of headland turnings where the operating with is less than twice the turning radius and for the case where the operating width exceeds twice the turning radius, respectively.

3. Results and discussion

As expected, the savings in non-working distance, and consequently in operational time, are increased as a function of increasing turning radius and as a function of decreasing operating width resulting in the same trend for all selected fields and for all conventional fieldwork patterns. This trend is indicated in the selected case of comparing *B-patterns* against AB-pattern in field REN that is presented in Fig. 6.

Table 1 provides the range of the savings in non-working distance and the range of the increases in area capacity for all fields and all comparisons of *B-patterns* against the three considered conventional patterns for all combinations between operating width and turning radius. It is shown that more complicated fields in terms of shape increases the average savings. The maximum savings in non-working distance occur in the case of field R41 when comparing *B-patterns* with *AB-patterns* while the maximum increase in area capacity occur in the case of BPL field when comparing *B-patterns* against *AB-patterns*. In total, the range of the savings in non-working distance was between 3.73% and 58.65% and the range of the increase in the area capacity was between 0.50% and 19.23%. These ranges indicate that the savings and increases in each case are strongly context dependent. Consequently, the derivation of overall average values is meaningless.

The comparison results are also presented graphically in Fig. 7, where Fig. 7a and b show the savings in non-working distance and the area capacity increase, respectively, for the selective case of comparing with *AB-patterns* and Fig. 7c and d regard the average savings in non-working distance and the average area capacity increase, respectively, for all patterns and field cases.

4. Conclusions

The assessment of the benefits of *B-patterns* showed savings in non-working distance up to 3.73% and 58.65% and increases in the area capacity up to 19.23% when compared with different types of conventional field-work patterns. This is expected to be minimum savings and increases based on the facts that a) the implementation of an exact algorithm (instead of the approximate algorithm used) can improve the optimality and the benefit of the solution in the case of large-scale problems (high number of tracks), b) physical fields are more complex that the simplified template fields used here, and c) only non-constrained operations in terms of capacity have been considered in the presented study.

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Appendices

Appendix I. Mathematical description of the optimisation problem

In a compact form, the optimisation problem underling the generation of $B^{*}_{\text{critetion}}$-patterns is of finding: $\sigma^{*} = \arg\min_{\sigma}[c^{criterion}_{0\ p^{-1}(1)} + \sum_{i=1}^{|T|-1} c^{criterion}_{p^{-1}(i+1),p^{-1}(i)} + c^{criterion}_{p^{-1}(|T|),f}]$, where $T = \{1, 2, 3, \ldots\}$ is the arbitrarily ordered set of the field tracks (indices) that cover the entire field area, $\sigma =< p^{-1}(1), p^{-1}(2), \ldots, p^{-1}(|T|) >$ is a permutation, $p^{-1}(\cdot) : T \rightarrow T$ is the inverse function of the bijective function $p(\cdot) : T \rightarrow T$ which for any field track $i \in T$, returns the position of the ith field track in the track traversal sequence in which the agricultural vehicle covers the field, $c^{criterion}_{0\ p^{-1}(1)}$ is the cost for the vehicle to move from its initial position (field entry point) to the first track in the traversal sequence, $c^{criterion}_{p^{-1}(|T|),f}$ is the cost for the vehicle to move from the end of the last track in the traversal sequence to its final position (field exit point), and $c^{criterion}_{p^{-1}(i+1),p^{-1}(i)}$ is the cost for moving between tracks $p^{-1}(i+1)$ and $p^{-1}(i)$.

Appendix II. Mathematical description of conventional field area coverage patterns

The mathematical descriptions of the conventional patterns considered in the presented assessment are as follows:

a. *AB-pattern*. For the case of these patterns, the function $p^{-1}(\cdot)$ is an identity function and thus $p^{-1}(i) = i, \forall i \in T$
b. *SF-pattern*. The sequence of the tracks is $\sigma = \langle 1, 3, 2, 5, 4, 7, 6, 8, 9, \ldots \rangle$. By excluding the first two traversed tracks, the rest of the pattern is a repetition of a standard motif of three tracks. Taking this into account, and also the possible irregularities of the motif at the last tracks, the traversal function can be written as:

$$p^{-1}(i) = \begin{cases} 1 & i = 1 \\ 3 & i = 2 \\ p^{-1}(i-1)+3 & \gamma(i) = 1 \wedge i \neq 2 \\ p^{-1}(i-1)-1 & \gamma(i) = 0, i \neq 1 \\ |T| & i = |T| \wedge \mathrm{mod}(|T|,3) \neq 0 \end{cases}$$

where "mod" represents the modulus operator.

c. *BL-patterns*. The sequence of the tracks is $\sigma = \langle 1,3,2,4,6,5,\ldots\rangle$. The patterns consist of the repetition of a standard motif of three patterns, that it can be written in a compact form as $p^{-1}(i) = i - 1 + \mathrm{mod}(i,3)$. Talking into account any irregularities in the motif at the last track, the traversal function is given by:

$$p^{-1}(i) = \begin{cases} i-1+\mathrm{mod}(i,3) & i \neq |T| \vee (i = |T| \wedge \mathrm{mod}(|T|,3) \neq 2) \\ |T| & i = |T| \wedge \mathrm{mod}(|T|,3) = 2 \end{cases}$$

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